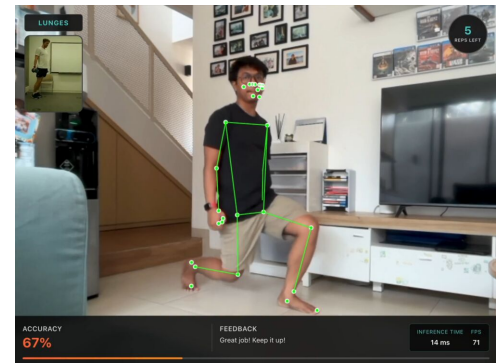


Portfolio — Solution Architecture & IoT Engineering. A selection of end-to-end IoT, embedded, and edge-AI solutions architected and delivered across industrial, defense, automotive, sports-tech, and consumer domains — spanning custom hardware, real-time firmware, cloud-scale backend systems, and mobile/web client applications. Each project demonstrates applied architecture and engineering across the full device-to-cloud stack, including embedded firmware, edge-AI inference, cloud-native backend systems, and mobile/web client applications.

1. GymSense — AI Fitness Tracker (XLabs, Indonesia)

Architected an AI-powered fitness tracking product as a flagship solution under XLabs' AI/IoT consultancy practice, demonstrating applied computer-vision and sensor-based movement analysis for real-time exercise tracking and form feedback.

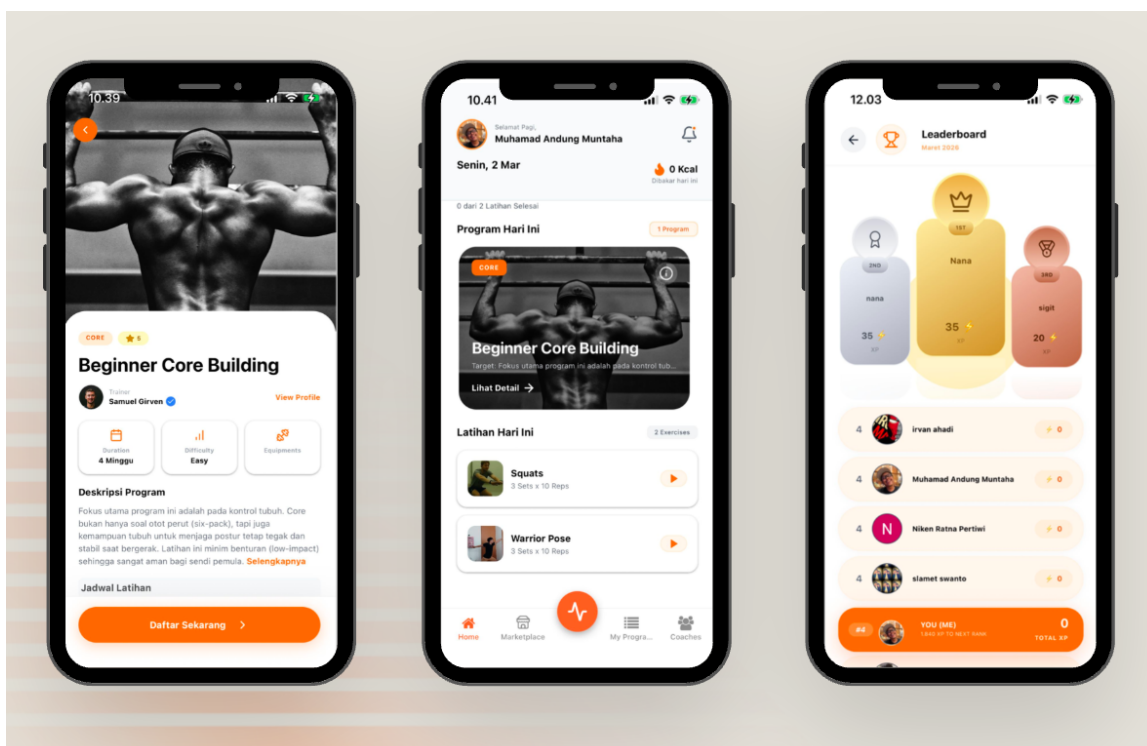
- Solution architecture spanning on-device inference, mobile/web client, and cloud backend
- Designed as a library to be integrated into third-party apps, enabling real-time exercise tracking and form feedback without requiring a dedicated device
- Live at: <https://xlabs.id/gymsense>



2. Movaja — AI-Coached Fitness App (XLabs, Indonesia)

Architected Movaja, a mobile and web fitness application combining pro-led training programs with real-time on-device AI form validation, giving users personal-trainer-level feedback without needing a gym or coach on-site. The platform spans a trainer/program marketplace, daily workout scheduling, and social leaderboard/gamification to drive engagement and retention.

- End-to-end mobile and web app architecture: program marketplace, daily workout tracking, progress/XP-based leaderboard, enterprise dashboard
- On-device AI processing for real-time exercise form scoring, enabling offline-friendly, low-latency coaching
- Live at: <https://movaja.id>



3. Secure Communication Device for Defense (Netra, Indonesia)

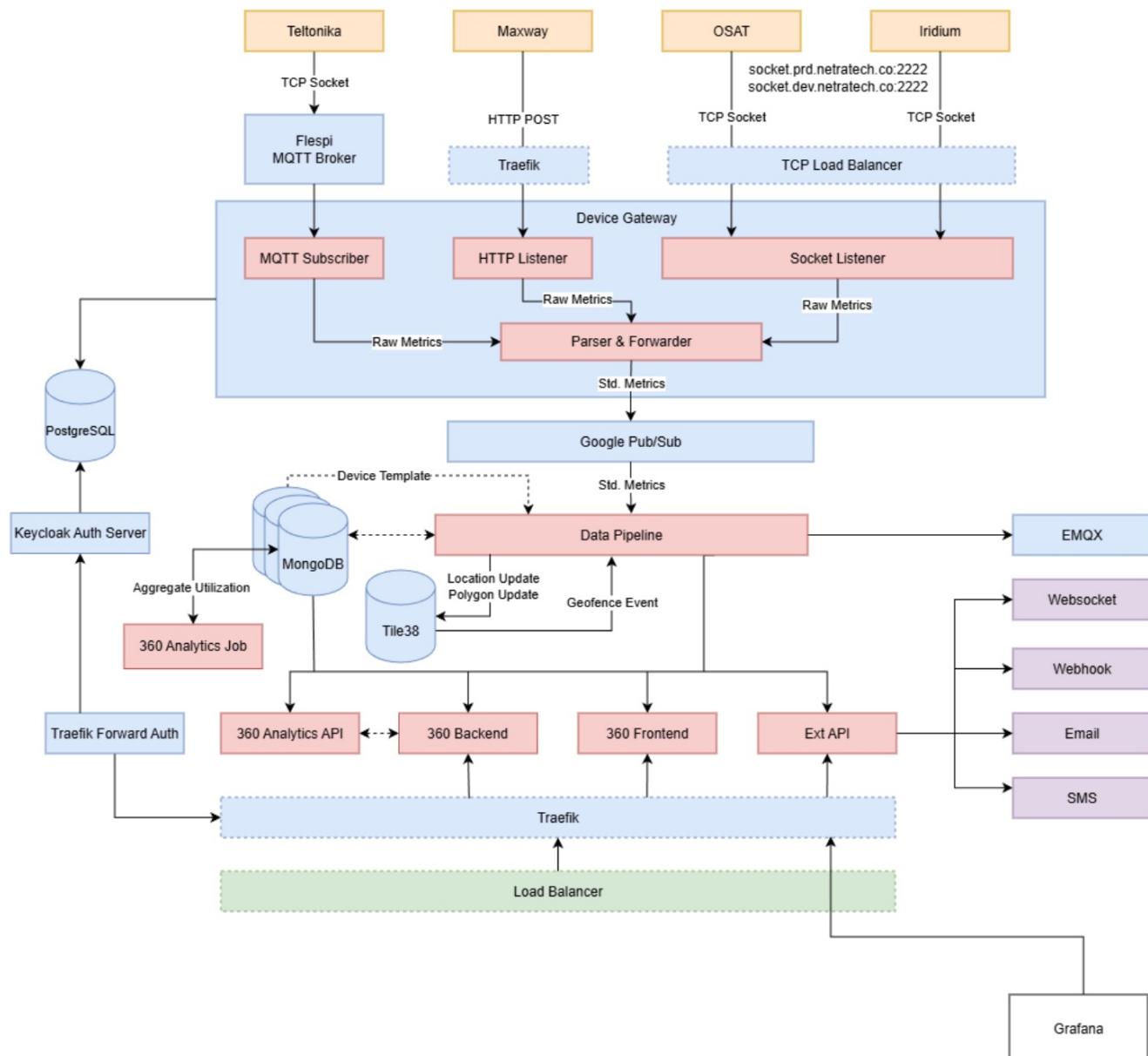
Led architecture and delivery of a custom embedded communication device for a defense-sector client, engineered for resilient field operation and secure data handling under mission-critical reliability requirements.

- Custom hardware and embedded firmware architecture built to defense-grade reliability standards
- Designed secure device-to-backend communication and remote device management
- Coordinated cross-disciplinary hardware, firmware, and systems integration teams



4. Enterprise IoT Platform (Netra, Indonesia)

Architected Netra's core IoT platform, connecting distributed embedded devices deployed on custom hardware to a Kubernetes-based cloud backend — covering device provisioning, telemetry ingestion, OTA firmware updates, and a client-facing management dashboard. Designed as the reusable architectural foundation behind multiple client engagements, prioritizing scalability, security, and maintainability across the full device-to-cloud stack.



5. Agora Gas Lift Optimization (SLB, Indonesia)

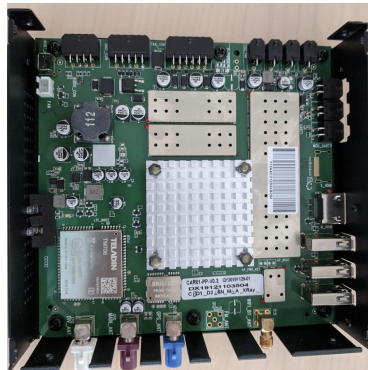
Architected and delivered Indonesia's first commercial Industrial IoT deployment: an edge-based, machine-learning-driven gas-lift optimization system built on SLB's Agora IIoT platform.

- Closed-loop control architecture continuously adjusting the gas injection control valve in real time
- ML-based optimization algorithm deployed and executed on the edge, integrated with SCADA
- Delivered a measured production increase of 5 BOPD, validating the solution's commercial impact



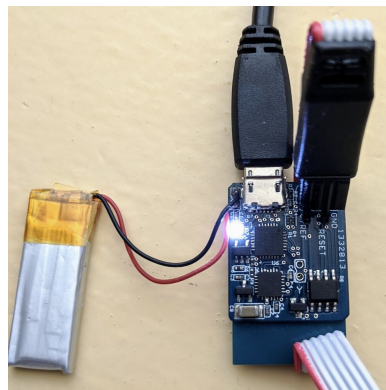
6. Fleet Metering Engine (Blue Bird Group, Indonesia)

Architected the real-time fare metering engine embedded within Blue Bird's fleet-wide IoT device, deployed across 20,000 vehicles. Integrated directly with each vehicle's CAN bus to derive accurate distance and fare calculations from live vehicle telemetry, feeding a device-management and OTA update platform running on a Kubernetes-based backend.



7. Wearable Sprint Tracker (SoleMetrix, Singapore)

Designed the end-to-end embedded architecture (Zephyr RTOS) for a sports-technology wearable, including a custom Ultra-Wideband (UWB) ranging protocol achieving 5cm positioning precision for tracking sprint speed and movement on the field. Defined the BLE peripheral API and device driver architecture, plus an efficient sensor-data compression scheme for high-frequency streaming to the host application.



8. Edge-AI Inference Engine (Nodeflux, Indonesia)

Architected the embedded Linux platform (Yocto-based) powering Nodeflux's edge-AI computing devices, including the OTA update strategy and remote-management design. Enabled CPU/GPU-accelerated machine learning inference at the edge for real-time video analytics, deployed in production across a fleet of edge devices.

